Human Nutrition

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్

भारतीय प्रौद्योगिकी संस्थान हैदराबाद

Indian Institute of Technology Hyderabad

What is Nutrition and why it is so important?

- The study of food, its composition, the amounts needed by the body and its effects on the body.
- At the most basic level, nutrition is about eating a regular, balanced diet.
- Good nutrition helps fuel your body. The foods you eat supply the nutrients your body needs to maintain your brain, muscle, bone, nerves, skin, blood circulation, and immune system.







- Proper nutrition also helps protect you from illness and disease, such as heart disease, diabetes, cancer, and osteoporosis.

- Nutrients are chemical substances required by the body to sustain basic functions and are optimally obtained by eating a balanced diet.

Six major classes of nutrients

- Carbohydrates
- Lipids/Fats
- Proteins
- Vitamins
- Minerals
- Water *(does not yield energy)*

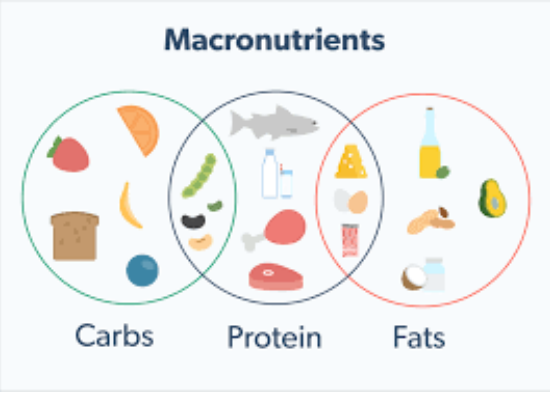
Major classes of nutrients in food

Macronutrients

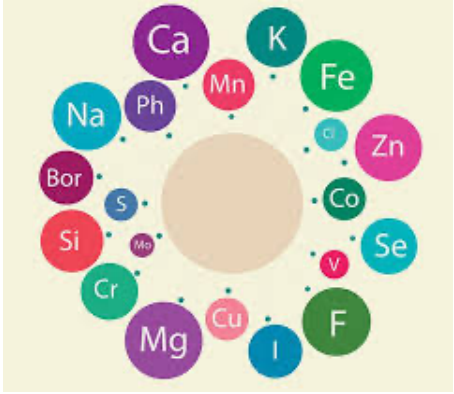
- *Includes:* Carbohydrates, Protein, and Fat *(produce energy)*
- supply energy (in the form of calories)
- serve as the building blocks for muscles and tissues

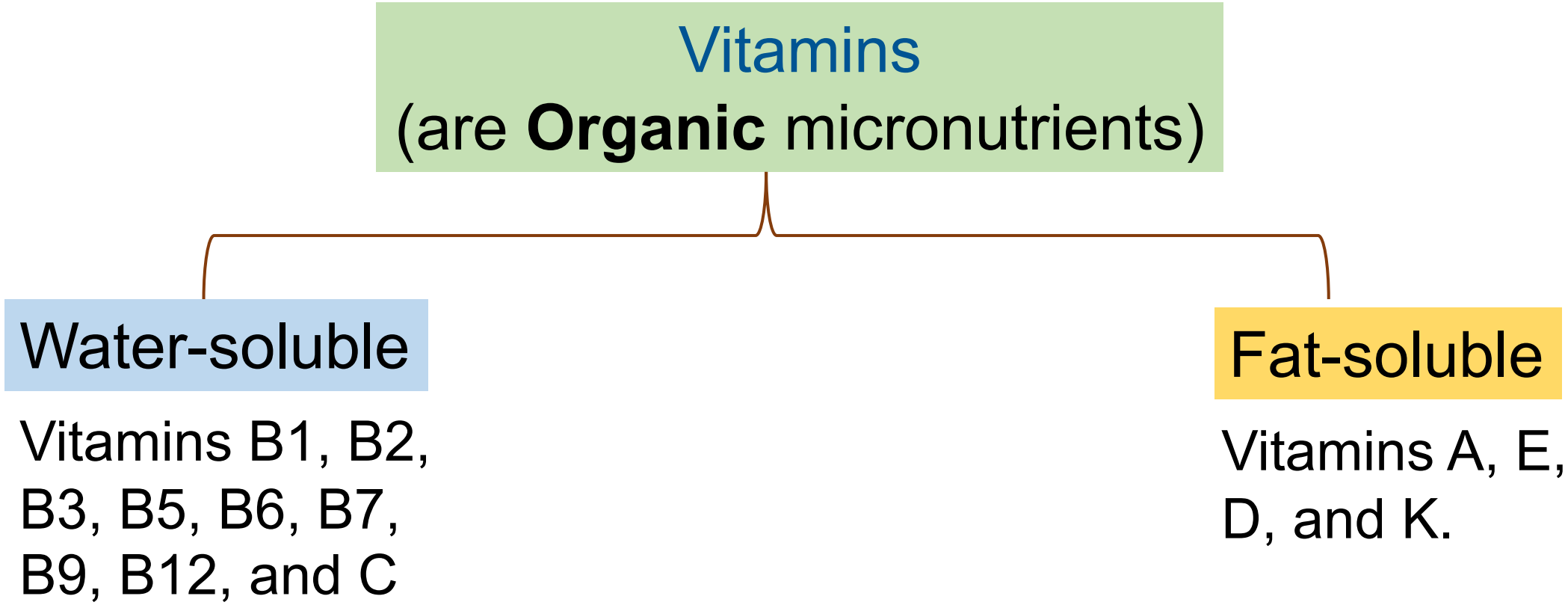
Micronutrients

- *Includes: Vitamins and Minerals (plays a role in metabolism)*
- Divided into four categories: water-soluble vitamins, fat-soluble vitamins, microminerals, and trace minerals








Vitamins
(are **Organic** micronutrients)
Water-soluble
Vitamins B1, B2, B3, B5, B6, B7, B9, B12, and C
Fat-soluble
Vitamins A, E, D, and K.

Minerals
(are **Inorganic** micronutrients)

Macrominerals

- Required in amounts greater than 100 mg per day
- Ex: Calcium, phosphorous, magnesium, sodium, potassium, and chloride.

Microminerals

- Required in amounts less than 100 mg per day
- Ex: Iron, copper, zinc, selenium, and iodine.

Carbohydrates:

- One of the essential macronutrients that are the primary source of energy for humans; 1 gram of carbohydrate contains 4 kcal of energy
- Carbohydrates are present in plant-based foods like grains, fruits, vegetables, and milk.
- Carbohydrates are ingested in the form of simple carbohydrates, like monosaccharides and disaccharides, or complex carbohydrates, like oligosaccharides and polysaccharides.
- Glucose (blood sugar) is the primary form to which carbohydrates become metabolized in humans.
- Lactose is a carbohydrate found in milk, and sucrose is basic table sugar.
- Fibers are plant carbohydrates (polysaccharides) like pectin and cellulose found in whole grains, fruits, vegetables, and legumes but are not digestible by humans.
- However, they play a major role in gut health and function and can be digested by microbiota in the large intestine.

There are three main types of carbohydrates

Sugars:

- They are also called simple carbohydrates because they are in the most basic form.
- They can be added to foods, such as the sugar in candy, desserts, processed foods, and regular soda.
- They also include the kinds of sugar that are found naturally in fruits, vegetables, and milk.

Starches:

- They are complex carbohydrates, which are made of lots of simple sugars strung together.
- Human body needs to break starches down into sugars to use them for energy.
- Starches include bread, cereal, and pasta.
- Also include certain vegetables, like potatoes, peas, and corn.

Fiber:

- It is also a complex carbohydrate.
- Human body cannot break down most fibers, so eating foods with fiber can help you feel full and make you less likely to overeat.
- Diets high in fiber have other health benefits. They may help prevent stomach or intestinal problems, such as constipation.
- They may also help lower cholesterol and blood sugar.
- Fiber is found in many foods that come from plants, including fruits, vegetables, nuts, seeds, beans, and whole grains.

Which foods have carbohydrates?

- Common foods with carbohydrates include:

Grains, such as bread, noodles, pasta, crackers, cereals, and rice
Fruits, such as apples, bananas, berries, mangoes, melons, and oranges
Dairy products, such as milk and yogurt
Legumes, including dried beans, lentils, and peas
Snack foods and sweets, such as cakes, cookies, candy, and other desserts
Juices, regular sodas, fruit drinks, sports drinks,energy drinks that contain sugar
Starchy vegetables, such as potatoes, corn, and peas







Which types of carbohydrates should I eat?

- When eating grains, choose mostly whole grains and not refined grains:
- Whole grains are foods like whole-wheat bread, brown rice, whole cornmeal, and oatmeal.
- They offer lots of nutrients that your body needs, like vitamins, minerals, and fiber.
- Refined grains are foods that have had some parts of the grains removed. This also removes some of the nutrients that are good for your health.
- Eat foods with lots of fiber.

- Try to avoid foods that have a lot of added sugar.
- These foods can have many calories but not much nutrition.
- Eating too much added sugar raises the blood sugar and can make gain weight.



How many carbohydrates one should eat?

- There is no one-size-fits-all amount of carbohydrates that people should eat. This amount can vary, depending on factors such as your age, sex, health, and whether or not one is trying to lose or gain weight.
- On average, people should get 45 to 65% of their calories from carbohydrates every day.
- On the Nutrition Facts labels, the Daily Value for total carbohydrates is 275 g per day.
- This is based on a 2,000-calorie daily diet.
- Daily Value may be higher or lower depending on your calorie needs and health.

Is it safe to eat a low-carb diet?

- Some people go on a low-carb diet to try to lose weight.
- This usually means eating between 25 g and 150 g of carbs each day.
- This kind of diet can be safe, but one should talk to the health care provider before starting it.
- One problem with low-carb diets is that they can limit the amount of fiber you get each day.
- They can also be hard to stay on for the long term.

Proteins

- Protein is in every cell in the body.
- Our bodies need protein from the foods we eat to build and maintain bones, muscles and skin.
- Proteins source: meat, dairy products, nuts, and certain grains and beans.
- Proteins from meat and other animal products are complete proteins. This means they supply all of the amino acids the body can't make on its own.



- Most plant proteins are incomplete.
- One should eat different types of plant proteins every day to get all of the amino acids your body needs.

- It is important to get enough dietary protein, Because human body doesn't store it the way it stores fats or carbohydrates.
- Protein requirement depends on your age, sex, health, and level of physical activity.

- 1 gram of protein contains 4 kcal of energy.

- The recommended protein intake is 0.8 to 1 gram per kilogram of body weight per days

For healthy children ages
1 to 3: 5 to 20%,
4 to 18: 10 to 30%,
Adults: 10 to 35%
of daily energy intake should come from protein, respectively.

Lipids/Fats

- Fat is a type of nutrient.
- Fats gives energy and help the body absorb vitamins.
- Lipids are essential macronutrients that are the main source of stored energy in the body, contribute to cellular structure and function, regulate temperature, and protect body organs.
- Lipids are found in fats, oils, meats, dairy, and plants and consumed mostly in the form of triglycerides
- One gram of fat contains 9 kcal of energy.



- But not all fats are the same. One should try to avoid:

Saturated fats such as butter, solid shortening, and lard

Trans fats.

- These are found in vegetable shortenings, some margarines, crackers, cookies, snack foods.

<u>For healthy children ages:</u>

1 to 3: 30 to 40%,

4 to 18: 25 to 15%,

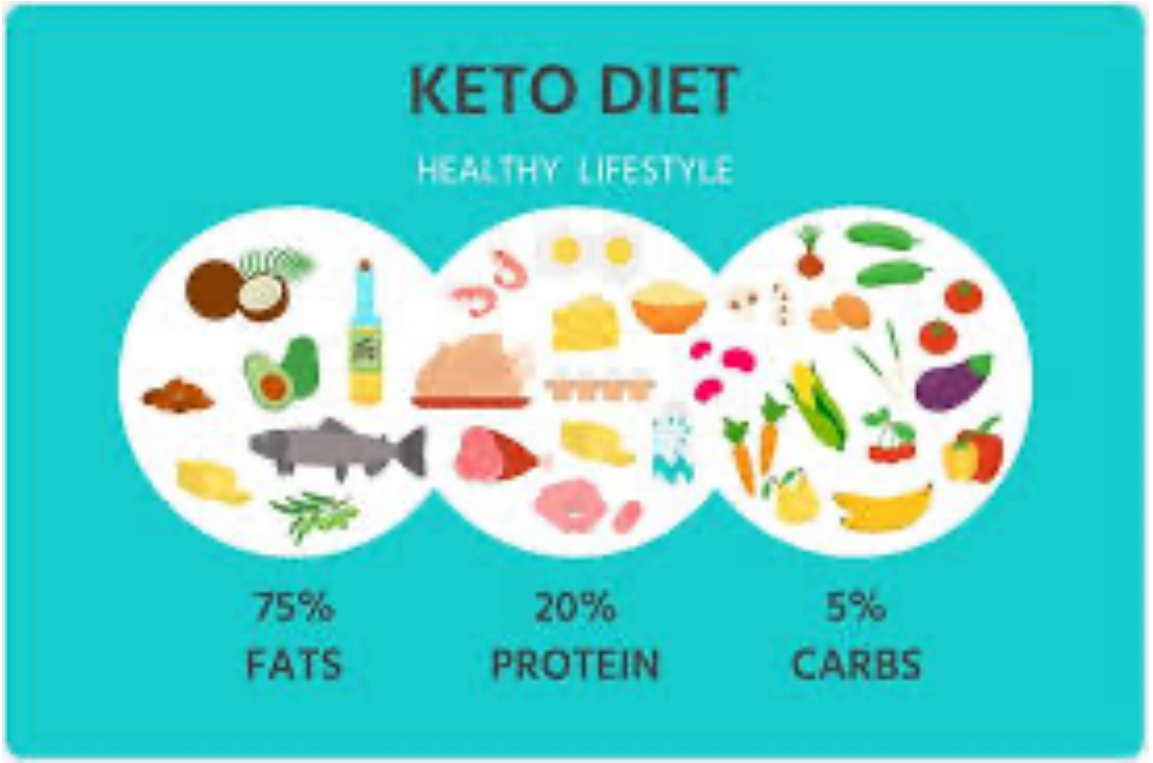
Adults: 20 to 35%

of daily energy intake should come from fat

Keto Diet

- The ketogenic diet is a very low carb, high fat diet

- It involves drastically reducing carbohydrate intake and replacing it with fat. This reduction in carbs puts your body into a metabolic state called ketosis.



Ketosis *is a metabolic process. When the body does not have enough glucose for energy, it burns stored fats instead. This results in a buildup of acids called ketones.*

- In keto diet; body becomes incredibly efficient at burning fat for energy.
- It also turns fat into ketones in the liver, which can supply energy for the brain

Is the keto diet healthy?

- Keto diet could reduce a person's risk of developing several health conditions, including metabolic syndrome, cardiovascular disease, and diabetes.

- These health benefits may occur due to the loss of excess weight and the inclusion of more nutritious foods in the diet, rather than the reduction in carbohydrates.

Vitamin B1 (Thiamin)

- Thiamin, or vitamin B1, is an essential water-soluble vitamin
- Acts as a coenzyme in carbohydrate and branched-chain amino acid metabolism.
- Thiamin is in food sources such as enriched and whole grains, legumes, and pork.
- The RDA (Recommended Dietary Allowance) of thiamin for adults is 1.1 mg/day for women and 1.2 mg/day for men.



Vitamin B2 (Riboflavin)

- Riboflavin, or vitamin B2, is an essential water-soluble vitamin
- Acts as a coenzyme in redox reactions
- Riboflavin is present in food sources such as enriched and whole grains, milk and dairy products, leafy vegetables, and beef.
- The RDA of riboflavin for adults is 1.1 mg/day for women and 1.3 mg/day for men.

Vitamin B3 (Niacin)

- Niacin, or vitamin B3, is an essential water-soluble vitamin
- Niacin is present in enriched and whole grains and high protein foods like meat, milk, and eggs.
- The RDA of niacin for adults is 14 mg/day of NEs (niacin equivalents) for women and 16 mg/day of NEs for men

Vitamin B12 (Cobalamin)

- Vitamin B12, or cobalamin, is an essential water-soluble vitamin Vitamin B12 is only present in animal products because it is a product of bacteria synthesis.
- Many foods are also fortified with synthetic vitamin B12.
- The RDA of vitamin B12 for adults is 2.4 mg/day.
- It is recommended for older adults to meet their RDA with fortified foods or supplements because many are unable to absorb naturally occurring vitamin B12.

Vitamin C (Ascorbic Acid)

- Vitamin C, or ascorbic acid, is an essential water-soluble vitamin
- Acts as a reducing agent in enzymatic reactions and nonenzymatically as a soluble antioxidant.
- Vitamin C is found primarily in fruits and vegetables, except for animal organs like the liver and kidneys.
- The RDA of vitamin C for adult women and men is 75 mg/day and 90 mg/day, respectively.

